Latihan 1

A. Selesaikan integral taktentu berikut.

1. $\int (x^4 - 3x^3 + 5x^2 - 11x + 4)\,dx = \frac{1}{5}x^5 - \frac{3}{4}x^4 + \frac{5}{3}x^3 - \frac{11}{2}x^2 + 4x + C$

2. $\int \left(4\sqrt[3]{x} - \frac{3}{\sqrt{x}}\right)dx = \int \left(4x^{\frac{1}{3}} - 3x^{-\frac{1}{2}}\right)dx$

$= \frac{4}{\frac{4}{3}}x^{\frac{4}{3}} - \frac{3}{\frac{1}{2}}x^{\frac{1}{2}} + C$

$= 3x^{\frac{4}{3}} - 6x^{\frac{1}{2}} + C$

3. $\int \left(\frac{5}{x^2} + \frac{4}{x^3} - \frac{6}{x^4}\right)dx = \int (5x^{-2} + 4x^{-3} - 6x^{-4})\,dx$

$= \frac{5}{-1}x^{-1} + \frac{4}{-2}x^{-2} - \frac{6}{-3}x^{-3} + C$

$= \frac{-5}{x} - \frac{2}{x^2} + \frac{2}{x^3} + C$

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4. $$\int \left(\frac{4x^2 - 3x - 1}{x^4} \right) dx = \int \left(4x^2 - 3x - 1 \right) \cdot x^{-4} \cdot dx$$

$$= \int \left(4x^{-2} - 3x^{-3} - x^{-4} \right) \cdot dx$$

$$= \frac{4}{-1} x^{-1} - \frac{3}{-2} x^{-2} - \frac{1}{-3} x^{-3} + C$$

$$= -\frac{4}{x} + \frac{3}{2x^2} + \frac{1}{3x^3} + C$$

5. $$\int \frac{t + 2t^2}{\sqrt{t}} dt = \int \left(t - 2t^2 \right) t^{-\frac{1}{2}} \cdot dt$$

$$= \int t^{\frac{1}{2}} - 2t^{\frac{3}{2}} \cdot dt$$

$$= \frac{1}{\frac{3}{2}} t^{\frac{3}{2}} + \frac{2}{\frac{5}{2}} t^{\frac{5}{2}} + C$$

$$= \frac{2}{3} t^{\frac{3}{2}} + \frac{4}{5} t^{\frac{5}{2}} + C$$

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B. Gunakan metode substitusi untuk menyelesaikan integral berikut.

6. $\int \frac{x^3}{(1+x^4)^2}\,dx$

Substitusi

$\Rightarrow u = 1 + x^4$

$\frac{du}{dx} = 4x^3$

$du = 4x^3 \cdot dx$

$\frac{du}{4} = x^3 \cdot dx$

$= \int \frac{\frac{du}{4}}{u^2}$

$= \int \frac{\frac{1}{4}}{u^2}\,du$

$= \int \frac{1}{4} u^{-2} \cdot du$

$= \frac{\frac{1}{4}}{-1} u^{-1} + C$

$= -\frac{1}{4u} + C$

$= -\frac{1}{4(1+x^4)} + C$

8. $\int x^2 \sqrt{x^3+1}\,dx$

Substitusi

$\Rightarrow u = x^3 + 1$

$\frac{du}{dx} = 3x^2$

$du = 3x^2 \cdot dx$

$\frac{du}{3} = x^2 \cdot dx$

$= \int \sqrt{u} \cdot \frac{du}{3}$

$= \frac{1}{3} \int u^{1/2} \cdot du$

$= \frac{1}{3}\left(\frac{2}{3} u^{3/2}\right) + C$

$= \frac{1}{3}\left(\frac{2}{3}(x^3+1)^{3/2}\right) + C$

$= \frac{2}{9}(x^3+1)^{3/2} + C$

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Turunan Fungsi Trigonometri

$y = \sin x \rightarrow y' = \cos x$

$y = \cos x \rightarrow y' = -\sin x$

$y = \tan x \rightarrow y' = \sec^2 x$

9. $\int \cos^2 x \sin x \, dx$

Substitusi

$\Rightarrow u = \cos x$

$\frac{du}{dx} = -\sin x$

$du = -\sin x \, dx$

$-du = \sin x \cdot dx$

$= \int u^2 \cdot (-du)$

$= -\int u^2 \cdot du$

$= -\frac{1}{3} u^3 + C$

$= -\frac{1}{3} \cos^3 x + C$

10. $\int \frac{\sin x}{\cos^2 x} dx$

Substitusi

$\Rightarrow u = \cos x$

$\frac{du}{dx} = -\sin x$

$du = -\sin x \, dx$

$-du = \sin x \cdot dx$

$= \int -\frac{1}{u^2} du$

$= -\int u^{-2} \cdot du$

$= -\frac{1}{-1} u^{-1} + C$

$= \frac{1}{u} + C$

$= \frac{1}{\cos x} + C$

$= \sec x + C$

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Latihan 5

1. Untuk soal berikut, tentukan $f(x)$

a) Diketahui $f'(x) = 6x^2 - 4x + 3$ dan $f(-1) = 3$

$$\int (6x^2 - 4x + 3)\, dx = \frac{6}{3}x^3 - \frac{4}{2}x^2 + 3x + C$$

$$= 2x^3 - 2x^2 + 3x + C$$

untuk mencari konstanta

$$f(-1) = 3$$

$$2x^3 - 2x^2 + 3x + C = 3$$

$$2(-1)^3 - 2(-1)^2 + 3(-1) + C = 3$$

$$-2 - 2 - 3 + C = 3$$

$$C = 3 + 2 + 2 + 3$$

$$C = 10$$

$$\therefore f(x) = 2x^3 - 2x^2 + 3x + 10$$

b) Diketahui $f'(x) = 6\sqrt{x} - \frac{10}{\sqrt{x}}$ dan $f(4) = 2$

$$\int 6\sqrt{x} - \frac{10}{\sqrt{x}}\, dx = \int 6x^{\frac{1}{2}} - 10x^{-\frac{1}{2}}\, dx$$

$$= \frac{6}{\frac{3}{2}}x^{\frac{3}{2}} - \frac{10}{\frac{1}{2}}x^{\frac{1}{2}} + C$$

$$= 4x^{\frac{3}{2}} - 20x^{\frac{1}{2}} + C$$

Mencari konstanta,

$$f(4) = 2$$

$$4x^{\frac{3}{2}} - 20x^{\frac{1}{2}} + C = 2$$

$$4(4)^{\frac{3}{2}} - 20(4)^{\frac{1}{2}} + C = 2$$

$$4(2^2)^{\frac{3}{2}} - 20(2^2)^{\frac{1}{2}} + C = 2$$

$$4(8) - 20(2) + C = 2$$

$$32 - 40 + C = 2$$

$$C = 2 - 32 + 40$$

$$C = 10$$

$$\therefore f(x) = 4^2\sqrt{x^3} - 20\sqrt{x} + 10$$

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c) Diketahui $f'(x) = 4\sin(2x) + 6\cos(3x)$ dan $f(0) = -4$

$$\int 4\sin(2x) + 6\cos(3x)\,dx = \int 4\sin(2x) + \int 6\cos(3x)\,dx$$
$$= 4\int \sin(2x) + 6\int \cos(3x)\,dx$$
$$= 4\cdot -\tfrac{1}{2}\cos(2x) + 6\cdot \tfrac{1}{3}\sin(3x) + C$$
$$= -2\cos(2x) + 2\sin(3x) + C$$

Mencari konstanta,

$$f(0) = -4$$
$$-2\cos(2x) + 2\sin(3x) + C = -4$$
$$-2\cos(2(0)) + 2\sin(3(0)) + C = -4$$
$$-2\cos(0) + 2\sin(0) + C = -4$$
$$-2(1) + 2(0) + C = -4$$
$$-2 + C = -4$$
$$C = -4 + 2$$
$$C = -2$$

$$\therefore f(x) = -2\cos(2x) + 2\sin(3x) - 2$$

d) Diketahui $f''(x) = 12x^2 - 6x + 2$, $f'(0) = 1$, dan $f(2) = 11$

$$f'(x) = \int 12x^2 - 6x + 2\,dx$$
$$= 4x^3 - 3x^2 + 2x + C_1$$

Mencari konstanta 1,

$$f'(0) = 1$$
$$4(0)^3 - 3(0)^2 + 2(0) + C_1 = 1$$
$$C_1 = 1$$

sehingga $f'(x)$

$$\therefore f'(x) = 4x^3 - 3x^2 + 2x + 1$$

⇒ Mengintegrasikan $f'(x)$ untuk mendapatkan $f(x)$

$$f(x) = \int 4x^3 - 3x^2 + 2x + 1\,dx$$
$$= x^4 - x^3 + x^2 + x + C_2$$

Mencari konstanta 2,

$$(2)^4 - (2)^3 + (2)^2 + 2 - 11 = -C_2$$
$$C_2 = -3$$

$$\therefore f(x) = x^4 - x^3 + x^2 + x - 3$$

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2. Gradien garis singgung pada kurva $y = f(x)$ di setiap titik (x, y) memenuhi bentuk $\frac{dy}{dx} = -9x + 3$. Jika kurva tersebut melalui titik $(-1, 3)$, tentukan persamaan kurva $f(x)$.

↳ $\frac{dy}{dx} = -9x + 3$

$dy = -9x + 3 \cdot dx$

maka persamaan kurva $y = \int -9x + 3 \cdot dx$

$= -2x^2 + 3x + C$

kurva melalui titik $(-1, 3)$

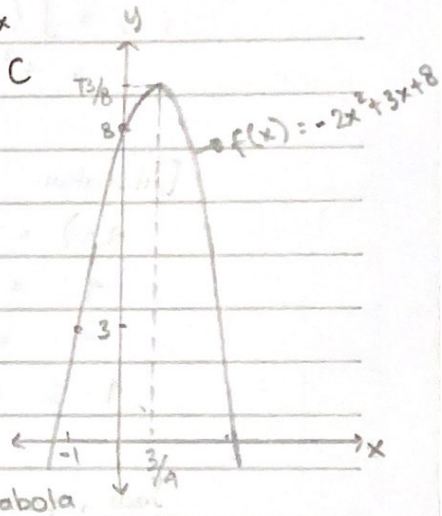
$y = -2x^2 + 3x + C$

$3 = -2(-1)^2 + 3(-1) + C$

$3 + 2 + 3 = C$

$8 = C$

$\therefore y = -2x^2 + 3x + 8$



⇒ Menemukan koordinat x dari titik puncak sebuah parabola

$f(x) = \underset{a}{-2x^2} + \underset{b}{3x} + 8$

$x = \frac{-b}{2a} = \frac{-3}{-4} = \frac{3}{4}$

Menemukan koordinat y ke dalam $f(x)$ substitusikan $x = \frac{3}{4}$

$f\left(\frac{3}{4}\right) = -2\left(\frac{3}{4}\right)^2 + 3\left(\frac{3}{4}\right) + 8$

$= -2\left(\frac{9}{16}\right) + \frac{9}{4} + 8$

$= -\frac{18}{16} + \frac{9}{4} + 8$

$= \frac{-18 + 36 + 128}{16}$

$= \frac{146}{16} = \frac{73}{8}$

$P\left(\frac{3}{4}, \frac{73}{8}\right)$

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3. Percepatan suatu benda bergerak pada setiap saat dinyatakan oleh $a(t) = (2t + 2)\ m/det^2$. Jika kecepatan pada $t = 1$ detik adalah $4\ m/det^2$, dan jarak yang ditempuh 6 meter. Tentukan rumus jarak s terhadap t !

↳ $a = \frac{dv}{dt}$

$$v(t) = \int a(t) \cdot dt$$
$$= \int (2t + 2) \cdot dt$$
$$= t^2 + 2t + C$$

Diketahui $v = 4\ m/det^2$ pada $t = 1$ detik

$$v(t) = t^2 + 2t + C$$
$$4 = (1)^2 + 2(1) + C$$
$$4 = 1 + 2 + C$$
$$4 - 3 = C$$
$$1 = C$$

Jadi, persamaan $v(t) = t^2 + 2t + 1$

$\Rightarrow$ $v = \frac{ds}{dt}$

$$s(t) = \int v(t) \cdot dt$$
$$= \int (t^2 + 2t + 1) \cdot dt$$
$$= \frac{1}{3}t^3 + t^2 + t + C$$

•) $t = 1$, $s = 6$ $\Rightarrow$ $6 = \frac{1}{3}(1)^3 + (1)^2 + 1 + C$

$$C = 6 - \frac{1}{3} - 1 - 1$$
$$= \frac{11}{3}$$

$$\therefore s(t) = \frac{1}{3}t^3 + t^2 + t + \frac{11}{3}$$

4. Suatu partikel bergerak sepanjang / sumbu x dengan kecepatan $v(t) = \sin 2t$. Jika posisi setiap saat adalah $x(t)$, tentukan $x(t)$ bila $x(0) = 0$

↳ $x(t) = \int \sin 2t \cdot dt$

Substitusi

$\Rightarrow u = 2t$

$du = 2 \cdot dt$

$dt = \frac{du}{2}$

$= \int \sin u \cdot \frac{dt}{2}$

$= \frac{1}{2} \int \sin u \cdot dt$

$= \frac{1}{2} - \cos u + C$

$= -\frac{1}{2} \cos 2t + C$

Mencari konstanta $x(0) = 0$

$x(0) = -\frac{1}{2} \cos 2(0) + C$

$0 = -\frac{1}{2} \cdot 1 + C$

$C = \frac{1}{2}$

Jadi, persamaan $\boxed{x(t) = -\frac{1}{2} \cos 2t + \frac{1}{2}}$

$x_{min} = -\frac{1}{2} \cos 2t + \frac{1}{2}$

$= -\frac{1}{2} \cdot 1 + \frac{1}{2}$

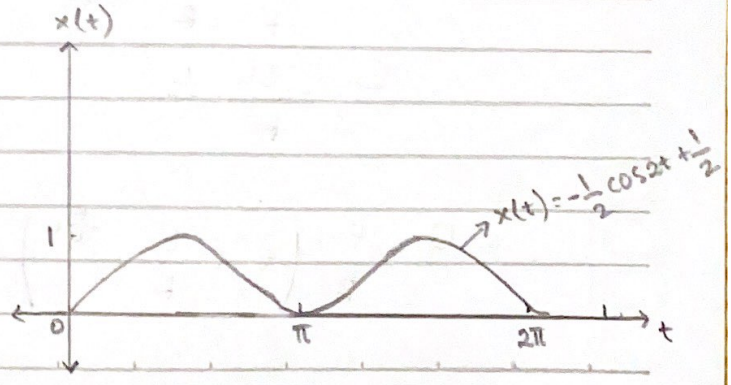
$= 0$

$x_{max} = -\frac{1}{2} \cos 2t + \frac{1}{2}$

$= -\frac{1}{2} \cdot (-1) + \frac{1}{2}$

$= 1$

$\cos 2t \rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$



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5. Arus dalam sebuah rangkaian listrik dengan induktansi L, tahanan R yang disusun seri dengan tegangan sumber konstan E memenuhi persamaan $i(t) = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$. Carilah persamaan muatan (q) untuk setiap t bila pada $t = 0$, muatan $q = 0$.

↳ $i = \frac{dQ}{dt}$

$$Q = \int i(t) \cdot dt$$

$$= \int \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right) \cdot dt$$

$$= \frac{E}{R} \int \left(1 - e^{-\frac{R}{L}t}\right) \cdot dt$$

$$= \frac{E}{R}\left(t - \frac{1}{-\frac{R}{L}} e^{-\frac{R}{L}t}\right) + C$$

$$= \frac{E}{R}\left(t + \frac{L}{R} e^{-\frac{R}{L}t}\right) + C$$

Persamaan muatan (q) bila $t = 0$, $q = 0$ yaitu :

$$Q(t) = \frac{E}{R}\left(t + \frac{L}{R} e^{-\frac{R}{L}t}\right) + C$$

$$0 = \frac{E}{R}\left(0 + \frac{L}{R} e^{-\frac{R}{L}(0)}\right) + C$$

$$0 = \frac{E}{R}\left(0 + \frac{L}{R} \cdot 1\right) + C$$

$$0 = \frac{E}{R} \cdot \frac{L}{R} + C$$

$$C = -\frac{EL}{R^2}$$

$$\therefore Q(t) = \frac{E}{R}\left(t + \frac{L}{R} e^{-\frac{R}{L}t}\right) - \frac{EL}{R^2}$$